

Home Bias or Systemic Risk? A Proposed Decomposition of Municipal Default Risk with Implications for Subsidies

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1 Proposal Summary and Related Literature

Municipal securities in the United States are well known to feature high credit spreads given their low observed probabilities of default. The observed cumulative 10-year default probability in the \$4 trillion municipal debt market is just .09%, compared to 2.23% in the \$10 trillion-plus corporate debt market.¹ Yet [Schwert \[2017\]](#) finds that risk premia for highly-rated municipal bonds are as much as 20 times higher than similarly rated corporates. Municipalities experience not only unusually high levels in their cost of capital, but also heightened volatility. [Haughwout et al. \[2024\]](#) show how the severity of spikes in municipal bond yields during the Covid-19 pandemic induced the creation of a new Federal Reserve program, the Municipal Liquidity Facility, indicative of particularly acute risk in the market.

Under standard arbitrage pricing theory, differences in credit spreads that run in the reverse direction of default probabilities imply that the risk profiles of investors and issuers in the municipal market follow a more cointegrated relationship than those in the corporate market. Recent literature has focused on home bias and imperfect risk-sharing, such as [Babina et al. \[2021\]](#) and [Pirinsky and Wang \[2011\]](#). Older literature like [Trzcinka \[1982\]](#) and [Fama \[1977\]](#) points to higher risks at a more systemic level. The relative importance of these two factors has important policy implications, given that municipal debt is free from state income tax for residents in most U.S. states, and research such as [Babina et al. \[2021\]](#) has shown the costs of market segmentation induced by this tax privilege.

In this proposal I explore the relative importance of investor home bias and systemic municipal credit risk through a model that augments [Babina et al. \[2021\]](#), and propose a research design to identify the relative importance of the two risk factors in municipalities across the U.S. The proposal seeks to provide further insight into the risk factors that are most salient to the representative investors of municipal bonds, and provide policy implications regarding tax privileges that most states offer.

The question of high risk premia in the market has been widely studied in the literature. Named the "muni puzzle", research from [Green \[1993\]](#), [Chalmers \[2015\]](#), and [Longstaff \[2009\]](#) investigated high yields for municipal bonds after adjusting for default probability. Past literature offered conflicting evidence on the whether the premia is primarily driven by

¹From 1970-2022, per the Municipal Securities Review Board.

liquidity, as offered by [Ang et al. \[2014\]](#)), or default risk, ([Schwert \[2017\]](#)).

Standing at nearly 40% the size of the entire US corporate debt market, the municipal debt market is the primary engine of investment in the various sectors that are largely under the purview of local governments, including education, safety, pension securities, and various forms of health and transportation infrastructure. Empirical evidence has repeatedly shown the cost of capital is puzzlingly high for these investments, enticing further research on the relevant discount factors for the representative investor in the market. For applied research on the costs of high premia, see [Haughwout et al. \[2024\]](#), [Goldsmith-Pinkham et al. \[2023\]](#), [Gao et al. \[2019\]](#), [Ivanov and Zimmermann \[2024\]](#), and [Ang and Green \[2011\]](#).

Following [Schwert \[2017\]](#) results that default risk explains between 74%-84% of the yield spread in municipal bonds from 1998-2015, I propose a decomposition of the relevant underlying risk factors that cause investors to discount payoffs from municipal debt, given its observed low default probability. Although Schwert convincingly demonstrates the importance of risk premia in explaining the spread, the underlying sources of the risk premia—particularly heterogeneity in the cross-section—are not well understood.

Recent research has focused on distortions created by the tax-exempt treatment of municipal bonds. Intuitively, the tax-exempt status is increasing in value with respect to an investor’s marginal income tax rate, and would be expected to encourage investor activity in higher income brackets, a prediction that is confirmed by [Bergstresser and Cohen \[2015\]](#) and [Ang and Green \[2011\]](#). [Ang & Green](#) and [Ivanov and Zimmermann \[2024\]](#) discuss segmentation at a national level, implied by the high share of municipal debt that is held by fund holdings and high net-worth retail investors—as much as 75% of all outstanding in 2020. [Babina et al. \[2021\]](#) takes these insights to show the importance of market segmentation in the state and local level in driving high risk premia, focusing primarily the effects of tax policy.

Older research on the municipal debt market has focused more the importance of systemic risk factors. [Trzcinka \[1982\]](#) notes that ratings agencies focus primarily on probability of default. As a result, municipal bonds may be riskier than corporate bonds with the same rating due to a number of factors, which including different bankruptcy proceedings or governance dynamics, as suggested by [Fama \[1977\]](#). [Gao et al. \[2019\]](#) touch on this idea by showing that the leniency of bankruptcy settlements is indeed an important contributor to default risk. [Haughwout et al. \[2024\]](#) note the potential importance of systemic credit risk in the market and propose a novel identification of municipal credit risk utilizing the Federal Reserve’s Municipal Liquidity Facility, but do not explore any decomposition of systemic and idiosyncratic risk.

This proposal will explore such a decomposition in a theoretical and empirical framework. I develop an augmented model of market segmentation following from [Babina et al. \[2021\]](#) that considers systemic municipal risk and evaluates the costs of tax privileges provided by state policy. I then propose an empirical design that exploits lending eligibility population cutoffs for The Municipal Liquidity Facility through a regression-discontinuity design to identify the relative importance of systemic and idiosyncratic risk in the municipal debt

market. The relative importance of these two factors has important implications for the optimality of state tax privileges, which will be explored in the model.

2 Theoretical Model

I present a model based on Merton [1987] that follows largely from Babina et al. [2021] with a few key differences. The model describes the price dynamics in the manner of a small open economy containing risk averse investors who can be local or global. There is a local municipal security and market security. Similarly to Babina et al. [2021], Local investors enjoy tax privileges from their state governments, decreasing the demanded tax-adjusted yield on the municipal security, but also carry the burden of local background risk that may be correlated with the idiosyncratic risk of that asset. The set of global investors is all those who have latent demand for the municipal security, but no tax privilege or correlated background risk.

The primary difference from Babina et al. [2021] is that I allow a flexible risk structure for global investors. I include a parameter for the share of idiosyncratic risk in the municipal security that global investors can hedge against by purchasing other municipal securities that are negatively correlated. The addition of this parameter has an implicit assumption of increased sophistication among the global investors, relative to local investors, a choice that is not trivial, but supported by the literature in municipal borrowing and investor home bias.² Secondly, the presence of additional, negatively correlated municipal securities is an important feature of the model in the background, but I only show the dynamics for one municipal security, along with the market security, for simplicity.

Like Babina et al. [2021], the model begins by solving for optimal asset demands of both types of investors and assumes market clearing to derive the expected equilibrium returns. I then investigate the dynamics of the demand curve and the net benefit of the tax privilege subsidy, showing comparative statics for the tax privilege and risk hedging parameter. The key insight of the model is that while the benefits of tax privileges can be self-defeating if they invite sub-optimal risk sharing, the strength of this relationship depends sharply on the ability of global investors to hedge against systemic municipal credit risk (Trzcinka [1982], Ang and Longstaff [2013]).

2.1 Securities and Investors

The asset profiles and objective functions follow from Babina et al. [2021]. The return on the municipal security is r_l and the return on the market security is r_m .

$$r_l = \bar{r}_l + \tilde{y} + \tilde{\epsilon}_l \quad (1)$$

$$r_m = \bar{r}_m + \tilde{y} \quad (2)$$

Where $\tilde{y}$ is a market wide systemic risk and $\tilde{\epsilon}_l$ is the idiosyncratic risk of the municipal security, $\tilde{Y} \sim (0, \sigma_y^2)$, $\tilde{\epsilon}_l \sim (0, \sigma_l^2)$, $Cov(\tilde{y}, \tilde{\epsilon}_l) = 0$. Both investors also have returns from background income and risk $\tilde{b} = \bar{b} + \tilde{\epsilon}_b$. Denote the local investors' background income as

²The implications of this assumption will be discussed in further detail later.

$\tilde{b}^l = \bar{b}^l + \epsilon_b^l$, and let $Corr(\tilde{\epsilon}_l, \tilde{\epsilon}_b^l) > 0$, $Corr(\tilde{\epsilon}_l, \tilde{\epsilon}_b^g) = 0$

All amounts thus far reflect post-tax amounts. For clarity, the model considers the impact of the distortion from a tax privilege, and uses the variable $\tau = \frac{Pre}{Post} - 1$ to denote the privilege $(1 + \tau)$ on an asset. The mean and variance of the total returns for the sets of investors are as follows

$$E[r^l + \tilde{b}^l] = \omega_l(1 + \tau)\bar{r}_l + (1 - \omega_l)r_m^- + \bar{b}^l \quad (3)$$

$$Var(r^l + \tilde{b}^l) = [\omega_l(1 + \tau)\sigma_l]^2 + [(1 + \omega_l\tau)\sigma_y]^2 + \sigma_b^2 + 2\omega_l(1 + \tau)\rho_l\sigma_l\sigma_b \quad (4)$$

$$E[r^g + \tilde{b}^g] = \omega_g\bar{r}_l + (1 - \omega_g)\bar{r}_m + \tilde{b}^g \quad (5)$$

$$Var(r^g + \tilde{b}^g) = \sigma_y^2 + \omega_g\rho_g\sigma_l^2 + \sigma_b^2 \quad (6)$$

Where r^l, r^g represent returns for local and global investors, respectively. $\omega_j \in [0, 1]$ is the weight that each investor group j places on the municipal security, as a share of their total wealth, given by W_j . $\rho_l \in [0, 1]$ is the correlation coefficient between the idiosyncratic risk of the municipal security and the local investors' background risk, while $\rho_g \in [0, 1]$ is the parameter that indicates the share of the municipal security's idiosyncratic risk that the global investor cannot hedge against in expectation. A natural way to interpret ρ_g is a common, or systemic component of municipal credit risk.

Following [Merton \[1987\]](#) and [Babina et al. \[2021\]](#), each investor j 's objective function is

$$\max_{\omega_j} E[r^j + \tilde{b}^j] - \frac{\delta}{2} Var(r^j + \tilde{b}^j) - \lambda_j \omega_l^j \quad (7)$$

Where δ is the investor's coefficient of constant relative risk aversion (assumed invariant throughout time and the cross-section) and λ_j is the Lagrange multiplier on the short sales constraint for the municipal asset (holdings of the municipal asset can not be negative).

The investors' optimal holdings, ω_l/ω_g can be found from first-order conditions, found by plugging in eq (3-6) into eq (7) and differentiating.

$$\omega_l = \frac{(1 + \tau)\bar{r}_l - r_m^- - \delta\tau\sigma_y^2 - (1 + \tau)\delta\rho_l\sigma_l\sigma_b}{\delta\gamma} \quad (8)$$

$$\omega_g = \frac{r_l - r_m - \lambda_g}{\delta\rho_g\sigma_l^2} \quad (9)$$

Where $\gamma = 1 + 2\tau + \tau^2(1 + \frac{\sigma_y^2}{\sigma_l^2})$

So far the model is identical to the model in [Babina et al. \[2021\]](#) apart from the expression of the tax privilege and the addition of the parameter ρ_g on the global investor's idiosyncratic risk. Market clearing will allow recovery of equilibrium expected returns in two distinct regions, as in [Babina et al. \[2021\]](#).

Let S be the total debt issuance by the municipality. Then the market clears with $S = \omega_l W_l + \omega_g W_g$. Where ω_l, ω_g are the share of portfolio allocation to the municipal security, and W_l, W_g are the total wealth of the groups of the investors.³

2.2 Region 1: $\frac{r_m^-}{1+\tau} + \frac{\delta\tau\sigma_y^2}{1+\tau} + \delta\rho_l\sigma_l\sigma_b < r_l < r_m$

When the non-privileged return on the asset does not reach that of the market security, the market is entirely populated by local investors, who are sensitive to correlated background risk and enjoy a lower intercept, but steeper slope on their demand curve for the municipal security. Market clearing gives

$$(1+\tau)r_l^l = r_m^- + \delta\tau\sigma_y^2 + (1+\tau)\delta\rho_l\sigma_l\sigma_b + \frac{\delta\gamma\sigma_l^2 S_1}{W_l} \quad (10)$$

$$r_l^l = \frac{r_m^-}{1+\tau} + \frac{\delta\tau\sigma_y^2}{1+\tau} + \delta\rho_l\sigma_l\sigma_b + \frac{\delta\gamma\sigma_l^2 S_1}{(1+\tau)W_l} \quad (11)$$

Where eq (11) represents the borrowing cost for the municipality, and the right side of eq (10) represents the total borrowing cost, including the loss in state tax revenue. S_1 represents the amount of debt that was issued under the conditions of region 1. The results in region 1 are again, identical to those in Babina et al. [2021]. The slope of the demand curve is increasing with both the idiosyncratic risk of the asset and the tax privilege given by the state. The intercept is lowered by the tax privilege, but shifted up by the correlation of the idiosyncratic risk with the investors' background risk. The effect of the correlated background risk is also not diminished by the amount of tax privilege, unlike the other components of the intercept.

2.3 Region 2: $r_m \leq r_l$

As the expected returns on the municipal security reach those of the market security, they will begin to attract global investors who do not receive a tax privilege. The sudden entrance of new investors with different risk profiles past a particular value creates a kink in the demand curve at that point. The correlated background risk and tax privilege still only apply to the local investors, but both groups of investors now share the idiosyncratic risk. Depending on the parameters ρ_l and ρ_g , the global investors may be better suited to holding the idiosyncratic risks of the asset. Market clearing recovers the following expression for the equilibrium returns in the region.

$$\left(\frac{(1+\tau)\rho_g W_l + \gamma W_g}{W_l + \gamma W_g} \right) r_l^g = \left(1 - \frac{(1-\rho_g)W_l}{W_l + \gamma W_g} \right) r_m + \left(\frac{W_l}{W_l + \gamma W_g} - \frac{(1-\rho_g)W_l}{W_l + \gamma W_g} \right) (\delta\tau\sigma_y^2 + (1+\tau)\delta\rho_l\sigma_l\sigma_b) + \frac{\delta\gamma\rho_g\sigma_l^2 S}{W_l + \gamma W_g} \quad (12)$$

³Given the assumptions the underpin the existence of the parameter ρ_g , it may be natural to view the municipal security not as a single security, but as a value-weighted average of all municipal securities in the investor's portfolio. Then ρ_g is the value-weighted correlation of risks across all of these securities. For simplicity, I assume the global investors' hedging against idiosyncratic risk is governed solely by ρ_g and do not model it otherwise.

Dividing out the coefficient on the left side then gives

$$r_l^g = (1 - \frac{\tau \rho_g W_l}{(1 + \tau) \rho_g W_l + \gamma W_g}) r_m + \left(\frac{\rho_g W_l}{(1 + \tau) \rho_g W_l + \gamma W_g} \right) (\delta \tau \sigma_y^2 + (1 + \tau) \delta \rho_l \sigma_l \sigma_B) + \frac{\delta \gamma \rho_g \sigma_l^2 S}{(1 + \tau) W_l (1 - (1 - \rho_g)) + \gamma W_g} \quad (13)$$

The demand curve after the kink point is similar to the portion before, but has a number of weights according to the wealth of local and global investors, which are further scaled by the tax privilege, τ , for local investors and the parameter for the systemic risk faced by global investors, ρ_g .

The intercept is shifted by the first two terms on the right, and the slope by the third term. The addition of these adjusted weights yields a number of new predictions. Firstly, as in Babina et al. [2021], as the wealth of the global investors goes to infinity, $W_g \rightarrow \infty$, all of the weights go to zero and we are left with $r_l = r_m$. When considering finite wealth, the parameter ρ_g plays an important role in determining the slope and intercept of the demand curve and is the primary theoretical contribution of this proposal.

At $\rho_g = 1$, the dynamics for the demand curve degenerate to the same as in Babina et al. [2021]. The intercept shifts up by a local wealth-weighted factor of the background and idiosyncratic risk covariance and the tax privilege. The slope is a function of the wealth-weighted shares of the tax-adjusted idiosyncratic risk. With $\tau = 0$ and $\rho_g = 1$, the slope is identical across regions and the kink disappears.

The overall cost of borrowing is clearly increasing with ρ_g , but it is important to note that ρ_g plays a key role in determining the slope of the demand curve in this second region. All else equal, higher ρ_g increases the slope of the demand curve, up to the limit of equality with the slope in the first region, absent the tax privilege. As ρ_g decreases, the slope tends closer to zero as global investors are able to enjoy the municipal security with a similar risk profile as that of the market. The importance of the predictions is explored in the next section, which provides comparative statics on the net benefit of the tax privilege.

2.4 Subsidy Cost and Comparative Statics

As mentioned before, the tax privilege is best thought of as a subsidy given from the state government to local governments to reduce their cost of borrowing. Although subsidies should not only be measured by net dollar benefit, some simple comparative statics will provide insight into dynamics that may affect the efficiency of such a subsidy. The importance of ρ_g should be clear, as it not only increases borrowing costs, it interacts meaningfully with the tax privilege variable.

Let S_l be the quantity of debt demanded by local investors in region 1 with a given tax privilege τ . Then the cost of the subsidy (to the state) is simply τr_l . Let the cost of the subsidy in region 1 be as in equation (11). The counterfactual cost of borrowing the same

amount, with no tax privilege is then equation (13), where $\tau = 0$. Then the net benefit ν of the subsidy is as follows

$$\begin{aligned} \nu = r_m \left(1 - \frac{1}{1 + \tau}\right) + (\delta \rho_l \sigma_l \sigma_b) & \left(\frac{\rho_g}{W_l + \gamma W_g} - \frac{1}{1 + \tau} \right) - \frac{\delta \tau \sigma_y^2}{1 + \tau} - \frac{\delta \gamma \sigma_l^2 S_l}{(1 + \tau) W_l} + \frac{\delta \rho_g \sigma_l^2 S_l}{\rho_g W_l + \gamma W_g} \\ & - \tau \left(\frac{r_m}{1 + \tau} + \frac{\delta \tau \sigma_y^2}{1 + \tau} + \delta \rho_l \sigma_l \sigma_b + \frac{\delta \gamma \sigma_l^2 S_1}{(1 + \tau) W_l} \right) \end{aligned} \quad (14)$$

Differentiating with respect to the supply, we obtain.

$$\frac{\partial \nu}{\partial \tilde{S}_l} = -\frac{\tau \delta \gamma \sigma_l^2}{(1 + \tau) W_l} + \delta \sigma_l^2 \left(\frac{\rho_g}{\rho_g W_l + \gamma W_g} - \frac{\gamma}{(1 + \tau) W_l} \right) \quad (15)$$

Eq (15) is the primary conclusion of the model, primarily regarding the second term, as the first term is simply the cost to the state. The second term reflects the rapidly diminishing benefit to the municipality from the tax privilege for its local investors. Importantly, the net benefit is higher when ρ_g is close to 1, which captures the fact that when the risk is highly systemic, the opportunity cost of attracting local investors over global ones is less severe. However when the term ρ_g is very low, then the opportunity cost is quite high as global investors are better suited to holding the idiosyncratic risk of the asset. Note that for supply of additional debt $S_g > S_l$, the net benefit would face similar dynamics in region 2, but weighted downwards by the wealth-weighted share of local investors. For simplicity, I illustrate the dynamics for region 1, where the majority of the cost is borne.

In the second term we also see the impact of reducing the price elasticity of demand for municipal debt through tax privileges, steepening the slope of the demand curve. The critical implication then is the higher the wealth-weighted share of local investors who are induced to purchase the local asset by the tax privilege, the steeper the (negative) relationship between debt issued by municipalities and the net benefit of the subsidy. Because the propensity of global investors to demand municipal debt is governed largely by ρ_g then the cost of the tax privilege policies for municipal debt depends highly on the share of municipal risk that is systemic, opposed to that which is purely idiosyncratic.

An important limitation of the model is that it assumes that global investors have a higher level of sophistication than the local investors, which allows them to hedge against some share of municipal default risk according to ρ_g . The inclusion of a similar parameter for local investors would likely reduce the costs of the home bias effect.

A natural extension of the model in its current form would add the effects of constraints on the ability or willingness of global investors to actually purchase municipal debt securities, which could result from financial markets being too localized or incomplete information problems such as in [Merton \[1987\]](#). Given the high share of local holdings documented by authors such as [Babina et al. \[2021\]](#) and [Bergstresser and Cohen \[2015\]](#), such explanations seem likely. Such additions would add further nuance to the importance of the parameter ρ_g , and could provide stronger justifications for tax privilege subsidy provided by state governments.

3 Data and Research Design

To explore the importance of the risk factors that are the focus of the model, I propose a research design that decomposes default risk components in the national U.S. market for municipal bonds. I will utilize the population eligibility cutoff of the Municipal Liquidity Facility in April 2020 to identify the idiosyncratic and systemic components.

Following [Schwert \[2017\]](#) and [Babina et al. \[2021\]](#), I consider only General Obligation (GO) bonds, purchases of which can be interpreted as a claim on the taxing power of the issuer and are not tied exclusively to any specific project, unlike Revenue bonds. I will obtain a comprehensive database of historical municipal debt issuance from 1998 onward from the Municipal Securities Rulemaking Board (MSRB), supplemented by additional characteristics from Bloomberg, Refinitiv, and Mergent. The MSRB dataset includes information on core characteristics such as CUSIP, date of trading and issuance, coupon rates, transaction price, and yield. Using the CUSIP identifier, each security can be matched to a richer set of characteristics in the various commercial databases, including tax-exempt status, the issuer name, principal amount, credit rating, coupon structure, callability, as well as information on pre-refunding.

Like Schwert, I guide my modeling of the risk premia by considering the decomposition of the tax-adjusted yield on all bonds into the sum of the risk-free rate (r_t), default risk (ϕ), and illiquidity (ψ) components.⁴ I use the interpolated maturity-matched swap rate as the risk-free rate. The simplest formulation is as follows

$$y_{i,t} = r_t + \phi_{i,t} + \psi_{i,t} \quad (16)$$

Where the default risk component encodes both the default probability and the expected loss given default. I will infer a weekly measure of the expected loss given default components (ϕ) of municipal debt from credit default swap (CDS) premium data from commercial data providers, such as Markit, Bloomberg, and S&P Global. The near equivalence of CDS premia and default risk components were shown by [Longstaff et al. \[2005\]](#). This validity of this choice is also supported by [Schwert \[2017\]](#) findings of similar decompositions using CDS premia and a liquidity and transaction approach. Following Schwert, I will obtain the default risk premium, defined as the implied default probability over the physical default probability for loss-given-default values of 25%, 50%, & 100%.

After obtaining estimates of default risk premia, I will exploit the eligibility cutoffs in the Municipal Liquidity Facility to identify the factors that comprise those premia. The Municipal Liquidity Facility was announced in April 2020 to provide much-needed sources of financing to states and municipalities that were having difficulty borrowing as the Covid-19 pandemic spiked yields and nearly froze the market entirely ([Haughwout et al. \[2024\]](#)). The program had eligibility requirements of minimum population thresholds, providing a clear continuous running variable with exogenous variation near the cutoff.

⁴Data for tax adjustments will come from the NBER-TAXISM model. I will separately estimate the regression without tax adjustments as a robustness exercise, using state-fixed effects to absorb tax privilege instead.

The Federal Reserve initially announced the existence of the program on April 9th, with a lending limit of \$500 billion and a minimum population eligibility of 1 million for cities, or 2 million for counties. Further expansions were announced on April 27th and June 3rd, reducing the population cutoffs to 250,000 and 500,000 respectively, and to include at least two cities or counties in each state.

Haughwout et al. [2024] use a regression-discontinuity design to identify the option value of the liquidity provision, and obtain an estimate of default risk utilizing a liquidity-default risk decomposition from Schwert [2017]. While Haughwout et al. [2024] identify the impact of the program on the eligible, or "treated" municipalities by looking at the differential effects between the right and left sides of the cutoff at the timing of the event, I am interested in the effects of the event on both sides of the cutoff.

As documented in Haughwout et al. [2024], the announcement of the Municipal Liquidity Facility had significant impacts on yields in the municipal debt market, regardless of eligibility, although they do find that eligible municipalities with lower credit ratings experienced more significant reductions. I propose modelling the announcement of the Municipal Liquidity Facility as the sum of two components. Firstly, as an event that reassured municipal investors, and decreased the salience of systemic credit risk across the entirety of the municipal market. The second component is the object of interest in Haughwout et al. [2024], the value of the liquidity provision itself for those who were eligible. The first component is analogous to the finding of Boyarchenko et al. [2022] that of market calming with the announcement of the Federal Reserve Corporate Credit Facilities during Covid-19.

The separation of the announcement into these two components lines up naturally with the two components of investor risk premia explored within the model. Although the Municipal Liquidity Facility did not remove either the systemic or otherwise idiosyncratic risk component entirely, it allows for a partial identification of each in an event-study context. The change in the default risk premium for municipalities on the left side of the cutoff reflects the effect of "calming" of perceived systemic risk attributable to the announcement of the facility itself. The change for municipalities on the right side is the sum of the "calming" effect and the liquidity provision that further mitigates idiosyncratic risk. The movement on the "left-side" of the cutoff has a clear interpretation as to whether ρ_g is close to 0 or 1, as it can only reflect a reduction in the systemic risk in the market. To extract insights from the "right-side" of the cutoff, ρ_g , I explore the heterogeneity in the "right side" of the change according to proxies of local investor home bias.

I propose a pooled regression discontinuity strategy, with the pre period spanning January 2020 - March 23 and the post-period spanning the same number of days (82) after announcements of eligibility criteria.⁵ The main specification will consider the post period as all trades following April 27th, following Haughwout et al. [2024]. To estimate the impacts of the facility separately on those eligible and those not eligible, I define a difference-in-differences regression discontinuity equation, with a dummy variable for the post period, a dummy for treatment, and a set of 2 polynomial slope coefficients for each period. To

⁵When estimating the effects of a particular announcement, the length will be adjusted downward on both sides to avoid including announcements that came later.

isolate the importance of home bias, I run the regression across stratified samples of proxies for the home bias. The regression equation will take the following form

$$P_{n(bi)}^h = \alpha + \delta \cdot \mathbb{I}(Post) + \tau \cdot \mathbb{I}(Pop \geq C)_i + \beta \cdot \mathbb{I}(Post)\mathbb{I}(Pop \geq C) + \gamma(\mathbf{Pop} - \mathbf{C}) + \mathbf{X}_{bi} + \epsilon_{n(bi)} \quad (17)$$

Where P represents the default risk premium recovered by methods described above, which is indexed by trade n of bond b for issuer i . δ is the change in intercept associated with the announcement of the facility, for both eligible and ineligible borrowers. τ is the coefficient on the dummy variable of eligibility based on the cutoff. β is the change in intercept associated with eligibility itself. γ is a vector of length 4 of polynomial slopes coefficients (2 for each period) for the centered running variable, population. Similar to [Haughwout et al. \[2024\]](#), I will define population as "relative population" according to the cutoff for eligibility, to be able to stack entities with different cutoffs together. $\mathbf{X}_{bi}$ is a vector of controls, which are not needed for identification but may improve precision in the estimates.⁶ The superscript h on $P_{n(bi)}^h$ is the indicator for the sample that the issuer i falls into. The samples will be stratified into quartiles of home bias proxies from [Babina et al. \[2021\]](#), tax privilege and the share of in-state fund holdings.

I will estimate the results using a triangular kernel, which assigns to diminishing weights to observations further from the cutoff. Bandwidth selection will follow the procedure outlined by [Imbens and Lemieux \[2008\]](#), which is a leave-one-out cross-validation approach. The coefficient δ , by design, can only reflect the impact of a reduction in systemic risk associated with the market-calming effect of the announcement.

To understand the relative importance of systemic opposed to purely idiosyncratic risk, as represented by ρ_g from the model, I note that the model predicts that the magnitude and significance of the coefficient δ provides evidence of the importance of systemic risk that was mediated by the announcement of the Municipal Liquidity Facility, and that it should vary minimally across stratified samples, denoted by h . The model predicts that the sensitivity (in absolute terms) of β to the level of home bias is decreasing in ρ_g . To fix ideas, suppose that $\rho_g = 0$, then β will be more strongly negative for samples with more home bias, as the level of systemic risk is minimal. If $\rho_g = 1$, then the risk is entirely systematic and the relationship with the home bias should be considerably weaker. Thus, the magnitude of the coefficient δ and the level of heterogeneity in the coefficient β across the stratified samples will provide suggestive evidence of the size of the parameter ρ_g .

4 Discussion & Broader Impacts

The proposal aims to provide a broader theoretical view of the determinants of credit risk in the municipal debt market, and model how the costs of tax privilege subsidies that are common in the United States may be governed by these determinants. The model's predictions

⁶I will not list all intended controls here, but one control of interest is the set of bankruptcy indicators from [Gao et. al \(2019\)](#). The change in the estimates following the inclusion of these controls will be further informative to the importance of the two risk factors.

suggest that the subsidy inefficiency attributed to tax privileges in the literature may be overstated if the systemic credit risk component is sufficiently large. With nuance however, the model also predicts that tax privileges may be particularly important in the presence of constraints to access by non-local investors, which may reduce the opportunity cost of such subsidies.

To assess these predictions, the proposal outlines a decomposition of municipal default risk that will point to the relative importance of the systemic and idiosyncratic risk components. Given the high levels of default risk premia in the municipal bond market documented by [Schwert \[2017\]](#)), municipalities likely have sub-optimal ability to make important investments in health, education, and infrastructure where they are responsible, as documented by many authors including [Goldsmith-Pinkham et al. \[2023\]](#) and [Ivanov et al. \[2022\]](#).

The empirical strategy of the proposal will provide estimates of the relative importance of the two risk factors that were mediated by the announcement of the Federal Reserve’s Municipal Liquidity Facility in April 2020. If the systemic risk factor is not particularly important, then the research will provide further evidence on the inefficiency of the tax privilege subsidy. On the other hand, if the systemic risk factor comprises a large share of the total default risk premium, then the research will suggest the subsidy’s inefficiency may be a response to other important factors in the market that limit the ability of sophisticated global investors to purchase a socially efficient level of municipal securities. The presence of such constraints, such as frictions in highly localized markets or asymmetric information problems are a promising area for even further research.

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